

AN AUTOMATED STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE WITH OPTIONAL JUMPS

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ABSTRACT. We specify a routine that calibrates the five canonical Heston [6] parameters $\Phi = (\theta, \kappa, \eta, \rho, v_0)$ to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot S_{ref} , the moneyness-normalised and grid-snapped strike K^* , the surface-coverage requirements, the configurable in-engine objective (defaulting to implied-volatility “IV-space” residuals), the IV-space restart selection and acceptance gate, and the boundary-pegging rejection rule. The same routine optionally fits the Bates [2] extension, which adds Merton lognormal jumps to the Heston dynamics, selected by a run-time flag and calibrated over the identical surface and gate. We keep Heston as the default throughout and present Bates as the eight-parameter generalisation it specialises from.

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1. The model

1.1. Basis. Under the risk-neutral measure, with constant risk-free rate r and continuous dividend yield g , the log-price $X_t = \ln S_t$ and instantaneous variance v_t of the SPX follow the Heston [6] dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where W, B are independent Brownian motions, so $\text{corr}(dX_t, dv_t) = \rho dt$. The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with $\theta > 0$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\eta > 0$ the volatility of variance (QuantLib's `sigma`), $\rho \in [-1, 1]$ the leverage correlation, and $v_0 > 0$ the initial variance. The Feller quantity [5]

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit; $\mathcal{F}(\Phi) \geq 0$ guarantees $v_t > 0$ almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

1.2. Jumps. The Bates [2] model extends Heston with Merton [8] lognormal jumps in the log-price. The variance dynamics (2) are unchanged. The level equation (1) gains a compound-Poisson jump term:

$$dX_t = (r - g - \frac{1}{2}v_t - \lambda\xi) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t) + dJ_t. \quad (5)$$

Here $J_t = \sum_{n=1}^{N_t} Y_n$ is a compound-Poisson process: N_t is a Poisson process of intensity $\lambda \geq 0$ (expected jumps per year), independent of W and B , and the log-jump sizes $Y_n \sim \mathcal{N}(\nu, \delta^2)$ are i.i.d. The compensator

$$\xi = \mathbb{E}[e^Y - 1] = e^{\nu + \delta^2/2} - 1 \quad (6)$$

offsets the jump drift so the discounted price stays a martingale, leaving the level drift $r - g$ in expectation. The augmented parameter vector is

$$\Theta = (\theta, \kappa, \eta, \rho, v_0, \lambda, \nu, \delta) \in \mathbb{R}^8, \quad (7)$$

the five Heston parameters of (1)–(2) plus jump intensity $\lambda \geq 0$, mean log-jump ν , and log-jump volatility $\delta > 0$. At $\lambda = 0$ the jump term vanishes and (5) collapses to the Heston level (1), so Heston is the $\lambda = 0$ face of the Bates parameter space. The model is selected at run time by a flag; the Heston fit is the default. The Feller quantity (4) is built from the diffusive parameters alone, so jumps do not enter it; it is reported under both models.

1.3. Pricing operators. Fix a valuation with maturity $\tau > 0$ (in years), strike K , and type $w \in \{+1, -1\}$ (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (8)$$

Black price. Black [3] and Black and Scholes [4], with total standard deviation $\Sigma = \sigma\sqrt{\tau}$,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (9)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where $\mathcal{N}$ is the standard normal CDF and $F = F(S, \tau)$.

Heston price. The Heston model price $C_H(\Phi; S, K, \tau, w)$ is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (10)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for $j \in \{1, 2\}$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (11)$$

and characteristic functions $f_j = \exp(C_j + D_j v_0 + iu \ln F)$ given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[(b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (12)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (13)$$

where $\mu_1 = \frac{1}{2}$, $\mu_2 = -\frac{1}{2}$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$. In code this integral is evaluated by the QuantLib [1] `AnalyticHestonEngine` (which uses a branch-stable formulation of (11)); equations (10)–(11) define the operator it realises.

Bates price. The Bates price $C_{\text{Bates}}(\Theta; S, K, \tau, w)$ has the same form as the Heston price (10)–(11), with each characteristic function f_j multiplied by the Merton lognormal jump factor of §1.2:

$$f_j^B(u) = f_j(u) e^{\tau \psi_j(u)}, \quad \psi_j(u) = \lambda \left(e^{iu_j \nu - \frac{1}{2} \delta^2 u_j^2} - 1 \right) - iu_j \lambda \xi, \quad (14)$$

with $u_2 = u$, $u_1 = u - i$ (the same measure shift that distinguishes f_1 from f_2), and $\xi = e^{\nu + \delta^2/2} - 1$ the compensator (6). The exponent ψ_j is the Merton log-jump cumulant, compensated by $-iu_j \lambda \xi$ so the drift correction of (5) carries through. At $\lambda = 0$ the factor is 1 and C_{Bates} reduces to C_H . In code the integral is evaluated by the QuantLib `BatesEngine`, the jump-augmented counterpart of the `AnalyticHestonEngine`.

Implied volatility. The Black-implied volatility of a price C is the inverse of (9) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (15)$$

computed numerically (QuantLib `blackFormulaImpliedStdDev` on the forward (8), returning Σ , then $\sigma = \Sigma/\sqrt{\tau}$). This operator defines the IV-space residuals of §5.

2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day d , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by i , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (16)$$

being respectively the quote timestamp, strike, type $w_i \in \{+1, -1\}$, traded size, trade price, trade-implied volatility, the underlying level *at trade time* t_i , and the days to maturity. Maturities enter the engine as τ_i in years under Actual/365, i.e. $T_i = \tau_i/365$.

The day is rate-stamped with $r = r(d)$ and $g = g(d)$, the most recent risk-free and dividend rates on or before d (a flat-forward curve for each; under flat curves the evaluation date is immaterial to T_i , so it is set to the QuantLib “today”).

Pre-filter. We keep only economically usable, Heston-appropriate quotes. Write the ratio moneyness $\varrho_i = e^{-|\ln(K_i/S_i)|} \in (0, 1]$ (it equals 1 at the money and falls as the strike moves out). The kept set is

$$\mathcal{D} = \{i : \sigma_i > 0, \tau_{\min} \leq \tau_i \leq \tau_{\max}, \underline{m} < \varrho_i < \bar{m}\}. \quad (17)$$

The maturity floor $\tau_{\min}$ (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive η, κ to Feller-violating extremes that pollute the pooled fit; the ceiling $\tau_{\max}$ drops thin, stale long tenors. The cutoff $\bar{m}$ removes the near-ATM rows, and the floor $\underline{m}$ removes the deep-OTM lottery-ticket tail ($\varrho_i < \underline{m}$, roughly $|\ln(K_i/S_i)| > 0.51$), whose extreme prices peg the fit to its bounds. Flooring that tail recovers acceptance and tightens the fit while keeping the full tradeable wing.

3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot S , yet the trades in $\mathcal{D}$ occur across an intraday range of underlying levels $\{S_i\}$. We reconcile the two with one reference spot and a moneyness re-striking.

Reference spot. Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (18)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH_MOVE} = \mathbf{1}_{\{R > R_{\max}\}}. \quad (19)$$

A flagged day ($R > R_{\max}$) is still calibrated and written, but the normalisation below is recorded as strained (its S_{ref} should be treated with suspicion).

Moneyness normalisation and grid snap. Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width Δ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (20)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in K/S rather than in the absolute strike. Re-striking by (20) therefore re-centres the whole day onto S_{ref} so that trades executed at

different S_i but the *same* moneyness share a common surface column. The grid snap collapses the continuum of K_i^* values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract’s *original* (S_i, K_i) , since the calibrated parameters are spot-independent.

4. Surface construction

From the normalised trades we build a partial $K^* \times \tau$ implied-volatility surface, selecting the most liquid quotes nearest the money. The trades entering this stage are already the pre-filtered set $\mathcal{D}$, so every candidate strike sits in the OTM moneyness band $\underline{m} < \varrho_i < \bar{m}$ of (17). That band fixes the reach of each wing: the cutoff $\bar{m}$ sets the inner edge, the strikes closest to the money, and the floor $\underline{m}$ sets the outer edge, where the deep-OTM tail is dropped. “Nearest the money” below therefore means just inside $\bar{m}$, not at the money itself.

Maturity selection. A maturity enters the surface only if it carries both wings. For each distinct maturity τ present in $\mathcal{D}$, let $n_c(\tau)$ and $n_p(\tau)$ be the number of distinct K^* on its call and put wings, and let $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$ be its traded volume. Call τ *admissible* when each wing carries at least $\underline{n}_k$ distinct strikes. Keep the $\bar{N}_t$ highest-volume admissible maturities,

$$\mathcal{T} = \text{top-}\bar{N}_t \text{ by } V(\tau) \text{ among } \{\tau : n_c(\tau) \geq \underline{n}_k, n_p(\tau) \geq \underline{n}_k\}. \quad (21)$$

The wing test precedes the volume cap. A maturity with a one-sided or single-strike wing is discarded before the cap, rather than consuming one of its $\bar{N}_t$ slots. The kept set falls below $\bar{N}_t$ only when fewer than $\bar{N}_t$ maturities are admissible.

Strike selection (per maturity, per wing). Within each kept maturity $\tau \in \mathcal{T}$, separate calls ($w_i = +1$, OTM with $K_i^* \gtrsim S_{\text{ref}}$) from puts ($w_i = -1$, OTM with $K_i^* \lesssim S_{\text{ref}}$) and keep the $\bar{n}_k$ nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{\bar{n}_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{\bar{n}_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (22)$$

The $\bar{n}_k$ cap binds only where a wing carries more than $\bar{n}_k$ distinct strikes inside the band; otherwise the wing’s outer edge is set by the floor $\underline{m}$ alone.

Pivot to a surface. For each retained cell (k, τ) with $k \in \mathcal{K}_\tau$, set the surface entry to the implied volatility of the *highest-volume* trade in that cell, the one with the largest traded size,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (23)$$

and leave it missing otherwise. When several trades tie at the maximum size the pick is arbitrary. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (24)$$

be the set of populated cells; these become the calibration targets (helpers).

Coverage requirement. To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (25)$$

A day failing (25) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

5. The optimization problem

The routine calibrates one parameter vector per day. Under the default Heston model this is Φ of (3) with $\dim \Phi = 5$; under Bates it is Θ of (7) with $\dim \Theta = 8$. The two are nested. They share their first $\dim \Phi$ coordinates, the diffusive parameters $(\theta, \kappa, \eta, \rho, v_0)$, and Θ appends the jump triple (λ, ν, δ) . Setting $\lambda = 0$ collapses the jump term (5) and recovers Φ , so Heston is the $\lambda = 0$ face of Bates. We write Ψ for the active vector, either Φ or Θ , and $C^{\text{mod}}(\Psi; \cdot)$ for the active model price: the Heston price C_H (10) when $\Psi = \Phi$, and the Bates price C_{Bates} (14) when $\Psi = \Theta$. Everything below is written in Ψ and holds for either model. The only place the model enters is the box dimension $\dim \Psi$ (26) and, through it, the bounds tables.

Let $\mathcal{N}$ index the targets with market data $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$, all priced at the single spot S_{ref} . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^{\dim \Psi} [\ell_j, h_j] \subset \mathbb{R}^{\dim \Psi}, \quad (26)$$

with (ℓ, h) in the `model.params()` order of Table 2: the five Heston coordinates, extended by the three jump bounds when $\Psi = \Theta$. The λ floor is exactly 0, so the Bates box contains the Heston model on its $\lambda = 0$ face.

5.1. Residuals. Two residual systems are defined: an IV-space system and a price-space system. The in-engine objective is configurable. It defaults to the IV-space system (28) below, and can instead be set to the price-space system (29). The choice changes only what each restart minimises. Selection (34) and the acceptance gate (36) always run on the IV-space error J_{iv} , so under the default objective the optimizer minimises the very error that then selects and gates.

IV-space (default in-engine objective; selection and gate). Invert each model price back to a Black vol via (15) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Psi) &= \mathcal{I}(C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Psi) &= \sigma_n^{\text{mod}}(\Psi) - \hat{\sigma}_n, \end{aligned} \quad (27)$$

with objective

$$J_{\text{iv}}(\Psi) = \left(\frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Psi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Psi) \text{ inverts finitely}\}. \quad (28)$$

J_{iv} is measured in volatility points; it is the natural surface-fit metric and, unlike J_{px} , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

Price-space (alternative in-engine objective). For each target the market value is the Black price at the quoted vol, $C_n^{\text{mkt}} = C_B(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$, and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Psi) = \frac{C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (29)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Psi) = \left(\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Psi)^2 \right)^{1/2}. \quad (30)$$

The per-target residual $\varepsilon_n^{\text{px}}$ is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt [7] is a sum-of-squares method, so it drives the squared residual norm $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Psi)^2$ to a local minimum (equivalently J_{px} , with which it shares a minimiser).

5.2. Multi-start local optimization. A single start into a non-convex problem is fragile [9], so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip} \left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau} \right)^2, [10^{-3}, 0.25] \right), \quad (31)$$

and define the seed set $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$ of $(v_0, \kappa, \theta, \eta, \rho)$ tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (32)$$

Under Bates each seed is extended by one fixed jump triple $(\lambda, \nu, \delta) = (0.1, -0.1, 0.1)$, so every restart starts from almost no jumps and grows them only if they tighten the fit; the restart count is unchanged. For each seed $s \in \mathcal{S}$ we solve the box-constrained nonlinear least-squares problem

$$\hat{\Psi}_s = \arg \min_{\Psi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{obj}}(\Psi)^2 \quad (\text{LM, started at } \Psi^{(0)} = s), \quad (33)$$

using Levenberg–Marquardt under the box constraint $\mathcal{B}$ (QuantLib `NonhomogeneousBoundaryConstraint`). The per-target residual $\varepsilon_n^{\text{obj}}$ is the configured in-engine objective: by default the IV-space residual $\varepsilon_n^{\text{iv}}$ (27), optionally the price-space residual $\varepsilon_n^{\text{px}}$ (29). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Psi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Psi}_s). \quad (34)$$

Ranking by J_{iv} (not J_{px}) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

6. Acceptance gate

Ψ^* is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction b_{tol} , be

$$\text{PEGGED}(\Psi) = \bigvee_{j=1}^{\dim \Phi} \left[(\Psi_j - \ell_j) < b_{\text{tol}} (h_j - \ell_j) \vee (h_j - \Psi_j) < b_{\text{tol}} (h_j - \ell_j) \right]. \quad (35)$$

The disjunction runs over the $\dim \Phi$ diffusive coordinates for both models. Under Bates the three jump coordinates (ν, δ, λ) are exempt from the predicate: $\lambda \rightarrow 0$ is a legitimate collapse to Heston rather than the optimizer hitting a wall, and when $\lambda \approx 0$ the jump shape (ν, δ) is unidentified and may rest on a bound without meaning. The jump triple is still reported. The Feller quantity (4)

likewise stays the diffusive quantity under Bates and never gates. The day is accepted iff

$$\boxed{J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Psi^*)}. \quad (36)$$

The first clause is the IV-space gate ($\varepsilon_{\text{iv}} \approx 2$ vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of J_{iv} . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

Why an IV-space gate. An earlier version gated on J_{px} directly. On a pooled day rich in deep-OTM wings, J_{px} is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with J_{iv} aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is ≈ 0.8 – 1.4 vol points, and the residual rejections are boundary-pegged κ or ρ (a separate identification lever), not fit-quality failures.

7. Output and diagnostics

Every attempted day produces output. Results are routed by model and objective into one directory, `results/⟨model⟩/calibrations/⟨objective⟩/`, so a Heston run never shares a file with a Bates run, nor an IV-space run with a price-space one. That directory holds three artefacts: a parameter table of accepted days (`calibrations.csv`), a rejection table of dropped days (`rejections.csv`), and a folder of per-day repricing diagnostics (`calibration_tests/`). The two tables partition the attempted days. An accepted day writes one row to the parameter table and nothing to the rejection table; a rejected day does the reverse.

Accepted days: the parameter table. An accepted day appends one row, keyed by the date d , to `calibrations.csv`. The row carries the rate-stamped inputs, the fitted vector, and fit and coverage diagnostics:

$$\begin{aligned} & (d; S_{\text{ref}}, r, g; \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*; \mathcal{F}(\Psi^*); \\ & J_{\text{iv}}(\Psi^*), J_{\text{px}}(\Psi^*), |\mathcal{N}|; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|, \sum_i q_i; \\ & \min_i S_i, \max_i S_i, R, \text{HIGH_MOVE}; t_{\text{calc}}). \end{aligned} \quad (37)$$

The blocks are: the reference spot (18) and the day's rates; the fitted parameters Ψ^* with their Feller quantity (4); the two objective values (28) (30) and the helper count $|\mathcal{N}|$; the surface coverage (kept maturities, distinct strikes, populated cells) and the day's total traded volume; the intraday spot range with its flag (19); and t_{calc} , the latest quote timestamp on the surface. A boolean accept flag (always true in this file) is stored alongside. Under Bates the jump triple $(\lambda^*, \nu^*, \delta^*)$ is inserted after v_0^* and nothing else changes. Because the calibration is once-daily, these rows accumulate into the one file, rewritten in date order at the end of each run.

Rejected days: the rejection table. A day dropped at any stage appends one row, also keyed by d , to `rejections.csv`, recording why:

$$(d; \text{REASON}, \text{DETAIL}; J_{\text{iv}}; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|). \quad (38)$$

The `REASON` is one of six categories, in pipeline order: `no_trades` (nothing survives the pre-filter (17)), `no_rate` (no r or g on or before d), `thin` (the coverage requirement (25) fails), `pegged` (the fit clears the IV gate but a diffusive parameter is boundary-pegged (35)), `iv_miss` ($J_{iv}(\Psi^*) > \varepsilon_{iv}$), and `no_fit` (no restart returned a finite fit). `DETAIL` is the human-readable string. The IV error and the coverage counts are filled in where the day reached the stage that computes them, and left blank otherwise: a `thin` day has coverage counts but no J_{iv} , a `no_trades` day has neither. The two tables together cover every attempted day, so the pegged-versus-thin-versus-miss split is auditable directly.

Per-day repricing diagnostics. Each accepted day also writes one file to `calibration_tests/`, holding the day's repriced surface contracts. One representative trade per populated cell (the highest-volume one, matching the pivot (23)) is valued under Ψ^* . The valuation uses each contract's *original* (S_i, K_i), not (S_{ref}, K^*): the structural parameters are spot-independent, and the honest comparison is against the actual trade. Each row records the original spot and strike, the snapped K^* , the fitted parameters, the market implied volatility σ_i , the Black price (9) at that vol, and the model price (the Heston price (10), or a Bates price column (14) under Bates).

8. Constants

Symbol	Code name	Value	Role
$\tau_{\min}$	MIN_DTM	14	minimum days to maturity
$\tau_{\max}$	MAX_DTM	730	maximum days to maturity
$\underline{m}$	OTM_MONEYNESSE_FLOOR	0.6	minimum OTM ratio moneyness
$\bar{m}$	OTM_MONEYNESSE_CUTOFF	0.98	maximum OTM ratio moneyness
$R_{\max}$	MAX_MOVE_PCT	0.03	intraday-range flag threshold
Δ	STRIKE_GRID	5.0	SPX strike lattice for K^* snap
$\bar{N}_t$	MAX_NT	20	maturities kept (by volume)
$\underline{n}_k$	MIN_NK	2	minimum distinct strikes per wing
$\bar{n}_k$	MAX_NK	40	strikes kept per wing
M_t	MIN_MATS	3	minimum maturities in surface
M_k	MIN_STRIKES	5	minimum strikes in surface
M_{cell}	MIN_CELLS	12	minimum populated cells $ \mathcal{N} $
ε_{iv}	IV_RMSE_ACCEPT	0.02	IV-space acceptance gate (vol pts)
b_{tol}	BOUND_TOL	10^{-3}	boundary-pegging tolerance fraction

TABLE 1. Surface, selection, and gate constants.

	θ	κ	η	ρ	v_0		λ	ν	δ
lower ℓ	10^{-4}	10^{-2}	10^{-2}	-0.999	10^{-4}	lower ℓ	0.0	-0.5	10^{-3}
upper h	1.0	20.0	2.0	0.5	1.0	upper h	5.0	0.2	0.5

TABLE 2. Box bounds $\mathcal{B}$ (26), in the `model.params()` order. Left: the five Heston coordinates ($\theta, \kappa, \eta, \rho, v_0$); the ρ upper bound is kept slightly positive (equity leverage implies $\rho < 0$) and η is capped at 2.0 (SPX vol-of-vol ~ 0.3 –1.2). Right: the Bates jump bounds (λ, ν, δ) appended when $\Psi = \Theta$; the λ floor is exactly 0 (the fit may collapse to pure Heston) and ν allows more negative room since equity jumps skew down.

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Appendix A. Algorithm

Algorithm 1 states the per-day routine in the active vector Ψ ; the driver applies it to every trading day and accumulates accepted rows into a single file (§7). Heston is the default ($\Psi = \Phi$). The Bates fit is the same algorithm with $\Psi = \Theta$: the engine is the **BatesEngine** and the box $\mathcal{B}$ is the dim Θ -dimensional one, while the pegging test (35) still ranges over the dim Φ diffusive coordinates only.

Algorithm 1 Per-day calibration

Require: day d of OTM trades; rates $r(d), g(d)$; constants of Table 1

- 1: $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$ (pre-filter (17))
- 2: **if** $r(d)$ or $g(d)$ undefined on/before d **then**
- 3: **return** $\emptyset$ (no rate)
- 4: **end if**
- 5: $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$;
 $R \leftarrow \max_i S_i / \min_i S_i - 1$;
 $\text{HIGH_MOVE} \leftarrow \mathbf{1}_{\{R > R_{\max}\}}$
- 6: $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$ with $m_i = K_i / S_i$ (re-strike (20))
- 7: $\mathcal{T} \leftarrow \text{top-}\bar{N}_t$ admissible maturities by volume ($\geq \underline{n}_k$ strikes per wing);
 build $\mathcal{K}_\tau$ (maximum $\bar{n}_k$ per wing);
 pivot to $\{\hat{\sigma}_{k,\tau}\}$, cells $\mathcal{N}$
- 8: **if** coverage (25) fails **then**
- 9: clear stale diagnostics; **return** $\emptyset$
- 10: **end if**
- 11: $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$; build seeds $\mathcal{S}$ (31) (32)
- 12: **for** $s \in \mathcal{S}$ **do**
- 13: $\hat{\Psi}_s \leftarrow \text{LM minimiser of } \sum_n (\varepsilon_n^{\text{obj}})^2 \text{ on } \mathcal{B} \text{ from } s$ (33)
- 14: compute $J_{\text{iv}}(\hat{\Psi}_s)$ (28)
- 15: **end for**
- 16: $\Psi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Psi}_s)$ (select (34))
- 17: **if** $J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}}$ **and** $\neg \text{PEGGED}(\Psi^*)$ **then**
- 18: write repriced diagnostics; **return** $\text{row}(d, S_{\text{ref}}, r, g, \Psi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20: clear stale diagnostics; **return** $\emptyset$ (rejected)
- 21: **end if**

Appendix B. Levers

The routine of §5–§6 carries three optional modifications, called levers. Each is a constant in the configuration (Table 3) and each is inert at the value zero, so the routine of the main text is recovered exactly when every lever is off. They act on two places only: the local problem (33) that each restart solves, and the rule (34) that picks one restart. None of them touches the acceptance gate (36), and none changes the reported error J_{iv} , so ε_{iv} keeps its meaning under any lever setting. The three levers are a wing weighting of the residuals, a soft Feller penalty on restart selection, and a cross-day parameter anchor. We define them in that order and then state the modified selection rule.

Wing weighting. The unweighted local problem treats every surface cell alike, so a body of near-money cells can outvote a sparse wing. The wing weight re-balances the cells by their distance from the money. For target n let

$$x_n = |\ln(K_n^*/S_{\text{ref}})| \quad (39)$$

be its absolute log-moneyness at the reference spot. The weight is

$$\omega_n = \max\left\{1 + \gamma (x_n/x_0)^\beta, \omega_{\min}\right\}, \quad (40)$$

with gain γ , ramp power β , reference log-moneyness x_0 , and floor $\omega_{\min}$. At the money $\omega_n = 1$. A positive gain raises the weight into both wings, reaching $1 + \gamma$ at $x_n = x_0$. A negative gain lowers it, so that near-linear days are not forced to bend the wings toward a few far cells; the floor keeps every weight strictly positive, which the optimizer requires. At $\gamma = 0$ every weight is one and the floor never binds. The weighted local problem replaces (33) by

$$\hat{\Psi}_s = \arg \min_{\Psi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \omega_n \varepsilon_n^{\text{obj}}(\Psi)^2. \quad (41)$$

The weights are normalised inside the optimizer, so only their ratios matter. The weighting is applied only under the IV-space objective (27). Under the price-space objective (29) the weights are held at one regardless of γ : the relative-price denominator already inflates the cheap wing cells, and a second weighting on top of it double-counts and drives ρ and η into their bounds. So that a weighted restart is ranked on the objective it minimised, the selection error becomes the weighted root mean square

$$J_{\text{iv}}^\omega(\Psi) = \left(\frac{\sum_{n \in \mathcal{N}'} \omega_n \varepsilon_n^{\text{iv}}(\Psi)^2}{\sum_{n \in \mathcal{N}'} \omega_n} \right)^{1/2}, \quad (42)$$

over the same finitely-inverting set $\mathcal{N}'$ as (28). The gate and the reported error stay the unweighted J_{iv} , and $J_{\text{iv}}^\omega = J_{\text{iv}}$ when the weights are uniform.

A sweep of the positive gain over $\gamma \in \{0.5, 1, 2\}$, at both narrow and wide strike coverage, was null. The lever engages (ρ drifts more negative, from about -0.57 to -0.60) but the residual by log-moneyness band does not improve at any band, and acceptance falls (from 64 to 44 of 100 days at $\gamma = 2$). Once the wings are inside the fitted surface there is no systematic wing bias left to weight away; what remains is the shape Heston can reach, not an objective-weighting problem. The negative-gain mode, intended for the over-convex call wing, has not yet been graded. The gain is therefore held at $\gamma = 0.0$.

Soft Feller penalty. The accepted set violates the Feller condition (4) on essentially every day. The violation is the driver of the elevated η , and through it of the non-monotone implied density that makes the fitted wing oscillate. A hard constraint is not an option: the short-tenor SPX surface genuinely asks for $\eta \approx 1$ with $\mathcal{F} < 0$, and forcing compliance destroys the fit. The lever instead biases the choice among restarts. Define the Feller shortfall

$$\mathcal{V}(\Psi) = \max\{0, \eta^2 - 2\kappa\theta\} = \max\{0, -\mathcal{F}(\Psi)\}, \quad (43)$$

which is zero on a compliant fit and grows with the size of the violation. Under Bates it is built from the diffusive coordinates alone, as $\mathcal{F}$ is. A restart is then charged $\mu_{\mathcal{F}} \mathcal{V}(\hat{\Psi}_s)$ on top of its selection error (equation (47) below), so

a compliant restart is preferred only when its extra fit cost is smaller than the penalty it saves. The selection error on an accepted day is about 0.005 and a typical shortfall is about 0.5, so a penalty near 0.01 makes the two terms comparable.

The penalty can only prefer a compliant restart if some restart converges into a compliant basin, and the six seeds of (32) do not: on calm days all six funnel into the same violating minimum. When $\mu_{\mathcal{F}} > 0$ the seed set is therefore extended by three Feller-compliant seeds, in the same $(v_0, \kappa, \theta, \eta, \rho)$ form as (32),

$$\mathcal{S}_{\mathcal{F}} = \begin{cases} (\hat{v}, 10.0, \hat{v}, 0.20, -0.70), \\ (\hat{v}, 6.0, 1.5\hat{v}, 0.15, -0.60), \\ (0.8\hat{v}, 8.0, 1.2\hat{v}, 0.25, -0.80), \end{cases} \quad (44)$$

each starting with a large κ and a small η so that $2\kappa\theta \geq \eta^2$ holds at the start. Under Bates each is extended by the same fixed jump triple as the base seeds. At $\mu_{\mathcal{F}} = 0$ the seed set is exactly $\mathcal{S}$ and the score is exactly J_{iv} .

Tested on a calm 2012 day under both models, the lever is weak. With the base seeds alone the penalty has nothing to choose between. With the compliant seeds and a strong penalty it selects a compliant restart whose error is about 0.057 vol points, far past the gate $\varepsilon_{iv} = 0.02$, and the day is rejected. Lowering the η cap merely moves the peg to the new ceiling. The penalty is thus a tie-breaker between near-equal restarts, not a cure, and it is held at $\mu_{\mathcal{F}} = 0.0$. The wing oscillation it was meant to address is treated instead through the accuracy of the characteristic-function integration in the pricing engine, which is a numerical setting rather than a change to the estimator.

Cross-day parameter anchor. Successive trading days share most of their surface, yet the day-by-day parameters can wander, most visibly the walk of κ toward its floor across a calm regime. The anchor damps this by softly regularising each day toward its own recent accepted history. Fix a day d and let $\mathcal{A}_d$ be the accepted days strictly earlier than d in the live accepted pool: the rows already on disk when the run started, plus every day the current run has accepted so far. If $\mathcal{A}_d$ is empty there is no anchor and the day is fitted without it. Otherwise take the last L days of $\mathcal{A}_d$ and set the anchor $\bar{\Psi}^{(d)}$ to their coordinate-wise median,

$$\bar{\Psi}_j^{(d)} = \text{median}\{ \Psi_j^{*(d')} : d' \in \text{the last } L \text{ days of } \mathcal{A}_d \}, \quad (45)$$

with $L = 1$ reducing to the single previous accepted day. The median over several days is a smoother prior than the last day alone and is robust to one odd fit. The anchor enters the engine in two places, both active only when $\mu_a > 0$ and an anchor exists. First, a warm-start seed at $\bar{\Psi}^{(d)}$ is appended to the seed set, so the basin of the recent history is always among the restarts; under Bates the warm seed is added only when the history carries the jump triple as well. Second, the selection score gains the span-normalised Tikhonov term

$$\mathcal{R}_a(\Psi; \bar{\Psi}^{(d)}) = \sum_j \left(\frac{\Psi_j - \bar{\Psi}_j^{(d)}}{h_j - \ell_j} \right)^2, \quad (46)$$

summed over the coordinates the anchor carries. Dividing by the box span puts every coordinate on a comparable 0–1 scale, so a κ move of a few units and a ρ

move of a few hundredths are charged alike relative to their admissible ranges. The term vanishes at the anchor and is weighted by μ_a in (47). A deviation of one full span costs μ_a against a selection error of about 0.005, so the weight is started small, in the range 0.001 to 0.01. On a well-identified day every restart lands near the anchor and the term is a near no-op, which is the property wanted: the anchor should not distort a good fit, and its bite is reserved for ill-conditioned days where several restarts tie on J_{iv} .

The anchor changes how the driver schedules days. Because day d anchors on its true latest predecessors, its fit depends on the outcome of day $d - 1$, which depends on $d - 2$, and so on. The chain is inherently sequential, and a day computed in parallel cannot anchor on a day still being fitted. When $\mu_a > 0$ the driver therefore processes the days strictly in date order on a single worker, and the parallel job count is ignored; an anchored run costs roughly the worker count times the wall-clock of the parallel run. At $\mu_a = 0$ no seed and no term are added and the parallel path is kept. On a resumed run the pool is seeded from the rows already on disk, so an incremental extension anchors on the days it continues from.

The modified selection rule. Write $\mathcal{S}^+$ for the seed set actually run: $\mathcal{S}$, extended by $\mathcal{S}_{\mathcal{F}}$ when $\mu_{\mathcal{F}} > 0$, and by the warm-start seed when the anchor is active. Each restart $s \in \mathcal{S}^+$ solves the weighted local problem (41) and is scored by

$$\mathcal{J}_{\text{sel}}(\widehat{\Psi}_s) = J_{iv}^{\omega}(\widehat{\Psi}_s) + \mu_{\mathcal{F}} \mathcal{V}(\widehat{\Psi}_s) + \mu_a \mathcal{R}_a(\widehat{\Psi}_s; \bar{\Psi}^{(d)}), \quad (47)$$

with the last term present only when an anchor exists. The selection rule (34) becomes

$$\Psi^* = \arg \min_{s \in \mathcal{S}^+} \mathcal{J}_{\text{sel}}(\widehat{\Psi}_s), \quad (48)$$

and a restart whose selection error is not finite is discarded before scoring. The acceptance test (36) is then applied to Ψ^* unchanged, on the unweighted $J_{iv}(\Psi^*)$ and the pegging predicate (35). In Algorithm 1 the levers change only the seed-building step, the restart loop, and the selection step: the seed set is $\mathcal{S}^+$, each restart minimises the weighted sum (41), and the selection ranks by $\mathcal{J}_{\text{sel}}$. With every lever at zero, $\mathcal{S}^+ = \mathcal{S}$, $\omega_n \equiv 1$, and $\mathcal{J}_{\text{sel}} = J_{iv}$, so (48) is (34) and the routine reproduces the main text to full floating-point precision. Table 3 lists the constants and their configured values; at the values shown the anchor is the only lever switched on.

symbol	config name	value	meaning
γ	WING_WEIGHT_GAIN	0.0	wing-weight gain; 0 = uniform weights
β	WING_WEIGHT_POWER	1.0	wing-weight ramp power
x_0	WING_WEIGHT_SCALE	0.05	log-moneyness where $\omega_n = 1 + \gamma$
$\omega_{\min}$	WING_WEIGHT_FLOOR	10^{-3}	floor on every weight
$\mu_{\mathcal{F}}$	FELLER_PENALTY	0.0	weight of the Feller shortfall (43)
μ_a	PARAM_ANCHOR_WEIGHT	0.005	weight of the anchor distance (46)
L	PARAM_ANCHOR_LOOKBACK	5	days in the anchor median (45)

TABLE 3. Lever constants. Each lever is inert at zero; the wing weighting also requires the IV-space objective.